

Elementary Problems Graph Theory

Version 1.0

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Basic Work

1. There are n people at a meeting. Some pairs of people shake hands with each other. Suppose the n people have shaken hands $d_1, d_2, \dots, d_n$ times respectively. Prove that the total number of handshakings is $\frac{d_1 + d_2 + \dots + d_n}{2}$.
2. There are $n \geq 3$ people at a meeting. Every pair of people may shake hands with each other or not. Suppose there is a person who does not shake hand with at least one other person. Find the maximum number of people who have shaken hands with everyone else.
3. (IMO shortlist 1966) There are $n \geq 2$ people at a meeting. Show that there exist two people at the meeting who have the same number of friends among the persons at the meeting. (It is assumed that if A is a friend of B , then B is a friend of A ; moreover, nobody is his own friend.)
4. There are n children. Some pairs of children are holding the two ends of some ropes such that each pair can hold at most one rope and each child may hold more than one rope. Suppose there are at least n ropes. Prove that we can find at least 3 children and arrange them in a circle such that any two adjacent children are holding the same rope.
5. There are n people at a meeting. Each person knows exactly 2 other people. Prove that we can arrange the people into some circles such that each person only knows the 2 neighbouring people in the same circle.
6. There are n cities in a country. Some pairs of cities are connected by flights. Suppose for each city, there are at least $\frac{n-1}{2}$ flights connecting this city and other cities. Prove that one can travel from any city to any other city through a number of flights.
7. There are some cities in a country. Some pairs of cities are connected by flights. One can travel from any city to any other city through a number of flights. Consider a flight between two particular cities A and B . Prove that the following are equivalent.
 - If the flight connecting A, B is cancelled, then one cannot travel from A to B through any number of flights.
 - One cannot form a cyclic trip by flights that includes the flight between A and B such that each flight is taken at most once and one can return to the starting point after the trip.
8. There are $n \geq 2$ people at a meeting. Suppose $n-1$ people know exactly $1, 2, \dots, n-1$ other people at the meeting respectively. How many people does the remaining person know?

9. (IMO shortlist 1967) In the space $n \geq 3$ points are given. Every pair of points determines some distance. Suppose all distances are different. Connect every point with the nearest point. Prove that it is impossible to obtain (closed) polygonal line in such a way.
10. There are at least 3 people at a meeting. Any two people either know each other or not. For any two people, they know exactly one common person. Prove that if two people do not know each other, then they know the same number of people.
11. There is an even number of points on the blackboard. Some segments are drawn which connect some pairs of points. Suppose for any two points, there is a path connecting them by travelling along the segments. Prove that we can erase some (possibly none) of the segments such that each of the remaining points is connected by an odd number of segments.
12. There are n people at a meeting. Some pairs of people are acquaintances. Prove that the following two conditions are equivalent.
 - The people can be partitioned into two groups such that any two people in the same group are not acquaintances.
 - There do not exist an odd number of people who can form a circle such that any two adjacent people are acquaintances.
13. (USAMO 1982 Problem 1) In a party with 1982 persons, among any group of four there is at least one person who knows each of the other three. What is the minimum number of people in the party who know everyone else?
14. (All-Soviet Union MO 1969) Every city in a certain state is directly connected by air with at most three other cities, such that one can get from any city to any other city with at most changing plane once. What is the maximum possible number of cities?
15. In a group of $n \geq 2$ people, a person is popular if the number of friends he knows in the group is strictly greater than the average number of friends that his friends have. Find the greatest number of popular people in the group.
16. There are some girls and boys in a room. Each girl knows at least 4 boys and each pair of girls know exactly one boy in common. Prove that we can distribute a black hat or a white hat to every boy such that each girl knows at least one boy with a black hat as well as one boy with a white hat.
17. (EGMO 2012 Problem 6) There are infinitely many people registered on the social network Mugbook. Some pairs of (different) users are registered as friends, but each person has only finitely many friends. Every user has at least one friend. (Friendship is symmetric; that is, if A is a friend of B , then B is a friend of A .)
 Each person is required to designate one of their friends as their best friend. If A designates B as her best friend, then (unfortunately) it does not follow that B necessarily

designates A as her best friend. Someone designated as a best friend is called a 1-best friend. More generally, if $n > 1$ is a positive integer, then a user is an n -best friend provided that they have been designated the best friend of someone who is an $(n-1)$ -best friend. Someone who is a k -best friend for every positive integer k is called popular.

- (a) Prove that every popular person is the best friend of a popular person.
 - (b) Show that if people can have infinitely many friends, then it is possible that a popular person is not the best friend of a popular person.
18. There are n people. Any two people may be friend of each other or not. Each person is a friend of 10 or 11 people. Every pair of people share exactly 5 common friends. Prove that $19 \leq n \leq 23$.
 19. There are $n \geq 3$ participants in a chess tournament. Each pair of participants play against each other exactly once and there is a unique winner in each play. Prove that there are at most $\frac{1}{4} \binom{n+1}{3}$ triples (A, B, C) of players such that A wins over B , B wins over C , and C wins over A .
 20. (Turkey MO 2nd Round 2014 Problem 6) 5 airway companies operate in a country consisting of 36 cities. Between any pair of cities exactly one company operates two way flights. If some air company operates between cities A, B and B, C we say that the triple A, B, C is properly-connected. Determine the largest possible value of k such that no matter how these flights are arranged there are at least k properly-connected triples.

Ramsey-typed Problems

21. (IMO shortlist 1969) Six points $P_1, \dots, P_6$ are given in 3-dimensional space such that no four of them lie in the same plane. Each of the line segments $P_j P_k$ is coloured black or white. Prove that there exists one triangle $P_j P_k P_\ell$ whose edges are of the same colour.
22. (Bosnia and Herzegovina MO 2011 Grade 11 Problem 4) Prove that among any 6 irrational numbers you can pick three numbers a, b and c such that numbers $a + b$, $b + c$ and $c + a$ are irrational.
23. (CGMO 2012 Problem 6) There are n cities ($n \geq 3$) and two airline companies in a country. Between any two cities, there is exactly one 2-way flight connecting them which is operated by one of the two companies. A female mathematician plans a travel route, so that it starts and ends at the same city, passes through at least two other cities, and each city in the route is visited once. She finds out that wherever she starts and whatever route she chooses, she must take flights of both companies. Find the maximum value of n .

24. (IMO 1964 Problem 4) Seventeen people correspond by mail with one another - each one with all the rest. In their letters only three different topics are discussed. Each pair of correspondents deals with only one of these topics. Prove that there are at least three people who write to each other about the same topic.
25. (USAMO 1978 Problem 5) Nine mathematicians meet at an international conference and discover that among any three of them, at least two speak a common language. If each of the mathematicians speak at most three languages, prove that there are at least three of the mathematicians who can speak the same language.
26. (CWMO 2005 Problem 8) There are n new students. Among any 3 people there are 2 of them who know each other. Among any 4 people there are 2 of them who do not know each other. Find the maximum value of n .
27. (CGMO 2006 Problem 4) 8 people join a meeting.
- (1) If for any 5 people there are 3 of them who are pairwise acquaintances, prove that there are 4 people who are pairwise acquaintances.
 - (2) If for any 6 people there are 3 of them who are pairwise acquaintances, is it always true that there are 4 people who are pairwise acquaintances?
28. (USAMO 1981 Problem 2) Every pair of communities in a county are linked directly by one mode of transportation: bus, train, or airplane. All three methods of transportation are used in the county with no community being serviced by all three modes and no three communities being linked pairwise by the same mode. Determine the largest number of communities in this county.
29. Prove the following upper bound for the Ramsey numbers. For any integers $m, n \geq 2$, we have

$$R(m, n) \leq \binom{m+n-2}{m-1}.$$

In other words, for any simple graph with $\binom{m+n-2}{m-1}$ vertices, there must be m vertices which are pairwise adjacent or n vertices which are pairwise non-adjacent.

Extremal Case

30. Each of the n people at a meeting knows at most 3 other people. Prove that we can partition all the people into two groups such that every person knows at most 1 person in the same group.

31. (USAMO 1989 Problem 2) The 20 members of a local tennis club have scheduled exactly 14 two-person games among themselves, with each member playing in at least one game. Prove that within this schedule there must be a set of 6 games with 12 distinct players.
32. There are some cities in a country. Some pairs of cities are connected by flights. One can travel from any city to any other city through a number of flights. Prove that the following are equivalent.
- For any city, there is an even number of flights connecting this city and other cities.
 - One can start a trip from a city by flights such that each flight is taken exactly once and one can return to the starting point after the trip.
33. Each of the n people at a meeting knows at least 3 other people. Prove that we can arrange an even number (≥ 4) of people in a circle such that every person knows both of his neighbours.
34. Each of the $2n$ people at a meeting knows at least n other people. Prove that we can arrange all the people in a circle such that every person knows both of his neighbours.
35. At a party with n people, it is known that among any 4 people, there are either 3 people who all know one another or 3 people none of which knows another. Show that the n people can be separated into two rooms, so that everyone in one room knows one another and no two people in the other room know each other.
36. At a party with $2n + 1$ people, it is known that for any group of n people, there exists a person not in this group who knows everyone in this group. Prove that there exists a person who knows all other people in the party.
37. (Argentina TST 2011 Problem 5) At least 3 players take part in a tennis tournament. Each participant plays exactly one match against each other participant. After the tournament has ended, we find out that each player has won at least one match. (There are no ties in tennis.)
- Show that in the tournament, there was at least one trio of players A, B, C such that A beat B , B beat C , and C beat A .
38. (Bulgaria MO 2004 Problem 3) A group consists of n tourists. Among every 3 of them there are 2 who are not familiar. For every partition of the tourists in 2 buses you can find 2 tourists that are in the same bus and are familiar with each other. Prove that there is a tourist familiar to at most $\frac{2}{5}n$ tourists.
39. (Brazil MO 2002 Problem 3) The squares of an $m \times n$ board are labelled from 1 to mn so that the squares labelled i and $i + 1$ always have a side in common. Show that for some k the squares k and $k + 3$ have a side in common.
- (Remark: We may assume $m, n \geq 2$ since otherwise the assertion is not true.)

40. (König's theorem) Let G be a bipartite graph. Let M be the maximum number of edges in G such that no two of them share a common vertex. Let N be the minimum number of vertices in G such that every edge in G is incident to one of these vertices. Prove that $M = N$.
41. (Vietnam TST 2001 Problem 3) Some club has 42 members. It is known that among 31 arbitrary club members, we can find one pair of a boy and a girl that they know each other. Show that from the club members we can choose 12 pairs of boys and girls knowing each other.

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